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INFINITY CONSULTANCY FIRM

BANGALORE, INDIA

Phase One Progress and Insights Report

Consultancy Report on Optimizing Sales Performance and Inventory Management at DL S Mall

For

DL S Mall Project Team

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Executive Summary

This consultancy report provides a comprehensive analysis of DL S Mall's sales performance, customer spending behaviour, and inventory management, to identify strategies to improve operations and foster future growth. The report examines sales data across different departments and floors, uncovering key patterns that can guide decision-making. The analysis highlights that the Ground Floor, particularly the Electronic Devices and Tech and Fragrances departments, generates the highest revenue. However, the Cinema Floor CF and Second Floor lag, showing areas that need improvement. A Pareto analysis shows that just seven departments contribute 80% of total sales, suggesting that the mall should focus on these high-performing sectors. A time-series analysis from 2020 to 2023 shows a solid recovery from the pandemic, with sales climbing from 2.16 billion AED in 2020 to 6.62 billion AED in 2023. Projections indicate continued growth, with sales potentially reaching 11.14 billion AED by 2026. This upward trend points to the need for effective resource allocation and strategic planning to capitalize on growth opportunities. For inventory management, the report's ABC analysis groups departments based on their contribution to sales. The top 22% of departments, which drive 76% of sales, should be the focus for inventory management and marketing. Meanwhile, lower-performing departments, which contribute only 4%, should be managed with tighter stock levels to reduce costs. The report provides recommendations to boost sales in underperforming areas, optimize inventory for top-selling departments, and ensure long-term growth through focused investments. In future, DL S Mall should assess customer traffic, marketing impact, and inventory turnover to fine-tune its strategy and improve profitability.

Introduction

The purpose of this consultancy report is to provide an in-depth analysis of DL S Mall's sales performance, customer spending patterns, and inventory management, to identify opportunities to optimize operations and drive future growth. This report includes a detailed examination of sales across different floors and departments, an analysis of key customer behaviors, and a review of the mall's inventory efficiency through techniques such as **Pareto** and **ABC analysis**. Additionally, the report highlights the impact of the COVID-19 pandemic on sales, offers forecasting insights for future growth, and provides actionable recommendations for improving underperforming areas and enhancing overall profitability.

Background

DL S Mall, a major shopping destination in Dubai, faced significant challenges during the COVID-19 pandemic, including reduced sales, store closures, and a shift toward online shopping. As the mall recovers, management is focusing on improving efficiency and customer satisfaction to stay competitive. Uneven sales performance and inventory issues have highlighted the need to reassess product categories and optimize space usage. As a consultancy, we will analyze sales data, identify top-performing areas, and provide recommendations for space and inventory optimization to support the mall's long-term growth and success in the post-pandemic market.

Methodology

The DLS Mall dataset, originally containing 11,715 entries, underwent a comprehensive data-cleaning process to ensure accuracy (Wolniak, 2023). During this process, 60 erroneous entries from the Net Sales Quantity column, including '0' values and one instance of '#VALUE!', were removed, as well as 25 records where the Gross Sales Amount was '0'. Text fields like Floor, Area, and Department were cleaned using Excel's TRIM() function to ensure consistency, while the Year and Month columns were merged into a single Date column in the format dd/mm/yyyy for easier analysis. Only valid numerical values greater than zero were retained in the Net Sales Quantity and Gross Sales Amount columns. Descriptive analytics techniques (Evans, 2016), including Pivot Tables (Grenci, 2022), were used to analyze sales by department, floor, and area, providing insights into performance. Pareto Analysis revealed that a small number of categories generated the majority of sales, following the 80/20 rule. Time-series analysis, using a line chart, tracked sales trends from 2020 to 2023, while heat maps highlighted high and low-performance areas. The ABC analysis prioritized products based on their contribution to revenue, calculating the Cumulative Gross Sales Percentage was calculated using the equation given below:-

$$\textbf{Cumulative Gross Sale Percentage} = \frac{\textbf{Cumulative Gross Sale}}{\textbf{Total Gross Sale Amount}} * \mathbf{100}$$

To classify departments A, B, and C for inventory management. These techniques provided a comprehensive understanding of sales distribution, customer behavior, and inventory optimization, aiding informed decision-making. Category A represents the top 20% of departments that generate 80% of revenue, Category B covers the next 30% contributing 15% of revenue, and Category C includes the remaining 50%, contributing just 5%. This method helps prioritize critical items for efficient inventory management. Forecasting uses historical data, such as Date and Gross Sale Amount, to predict future sales trends using models like trendlines. This enables businesses to make informed decisions, plan for growth, and optimize strategies for long-term success based on potential future performance.

RESULTS

Sales Performance Analysis

Gross Sales by Top 5 Departments Across Floors and Areas

The analysis of total gross sales across various wings and floors highlights the dominance of the Ground Floor in DL S Mall as shown in Figure 1.

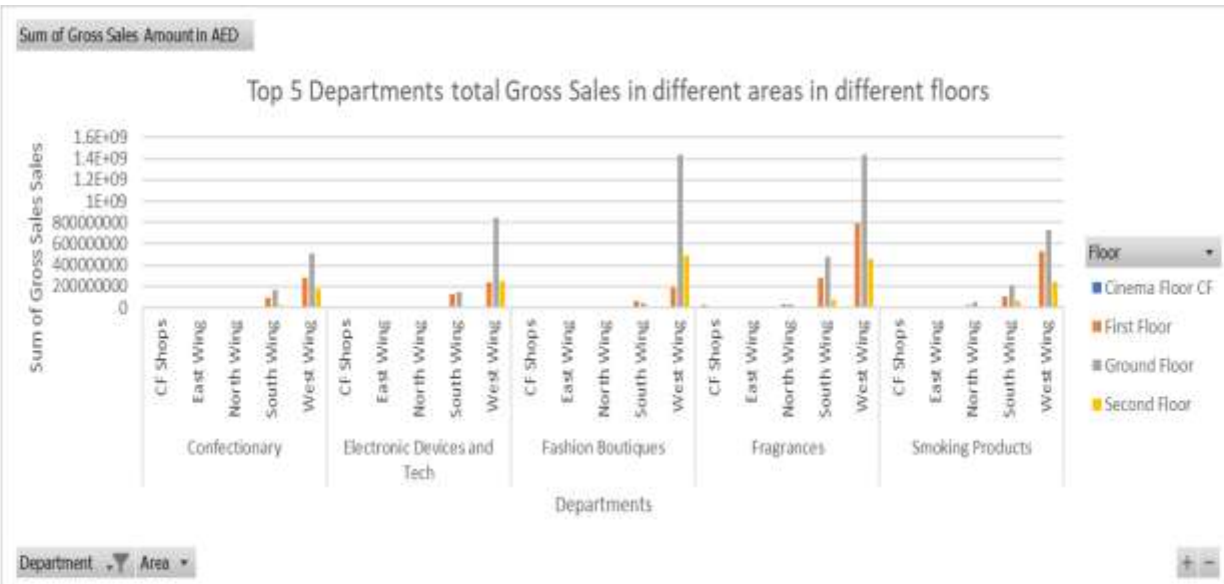

Figure 1. Total Gross Sales by the Top 5 departments in different areas and floors

Figure 1 shows that Electronic Devices and Tech have the highest sales, particularly in the West Wing, while Fashion Boutiques and Fragrances also perform well, especially in the East Wing and CF Shops. In contrast, Confectionary and Smoking Products have lower and more evenly distributed sales across various wings and floors. The graph highlights how sales performance varies by department and floor, with some departments performing better on specific floors like the Cinema Floor CF and Ground Floor.

Pareto Analysis

The Pareto analysis in Figure 2 highlights that 7 key departments, including Fragrances, Fashion Boutiques, and Smoking Products, account for 80% of the total gross sales in DLS mall, demonstrating the 80/20 rule.

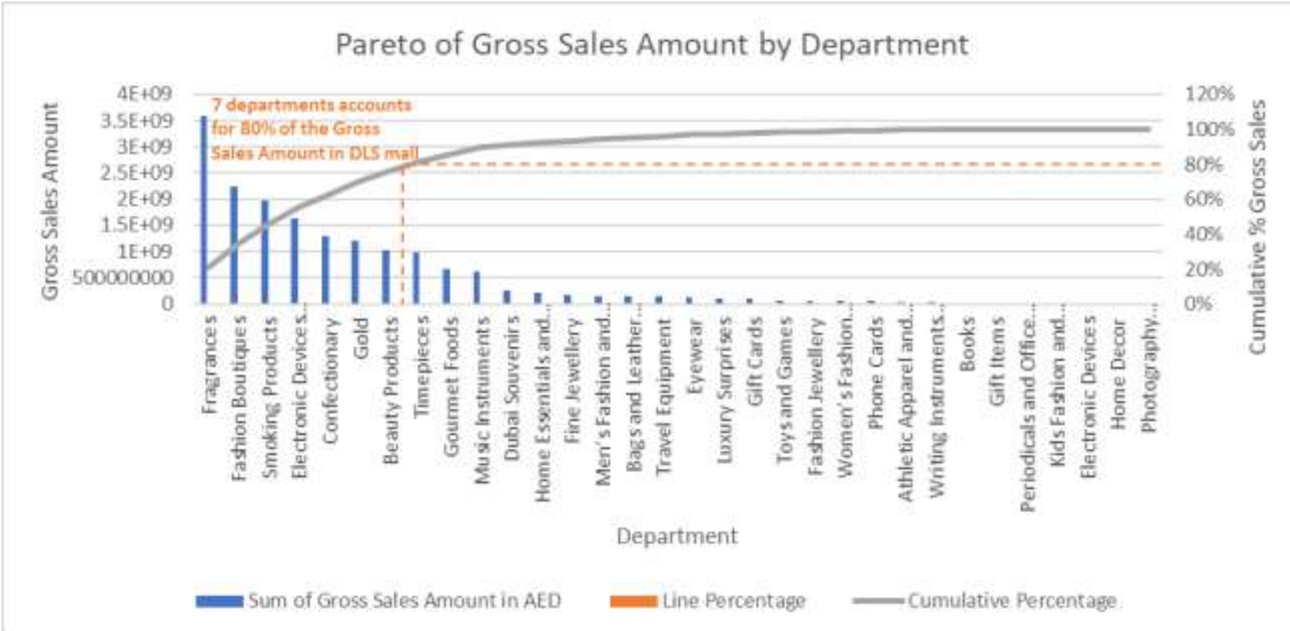

Figure 2. Pareto Analysis

The blue bars show that these departments dominate sales, while the remaining departments contribute much less. The cumulative percentage (grey line) rises steeply for the top departments and flattens out for the rest, showing their smaller impact. To maximize impact, the mall should focus on optimizing performance in these top 7 departments, while exploring ways to enhance or streamline the lower-performing ones.

Customer Spending Patterns

Time-Series Analysis

This time series analysis as shown in Figure 3 of gross sales from 2020 to 2023 at DL S Mall shows a steady upward trend in customer spending.

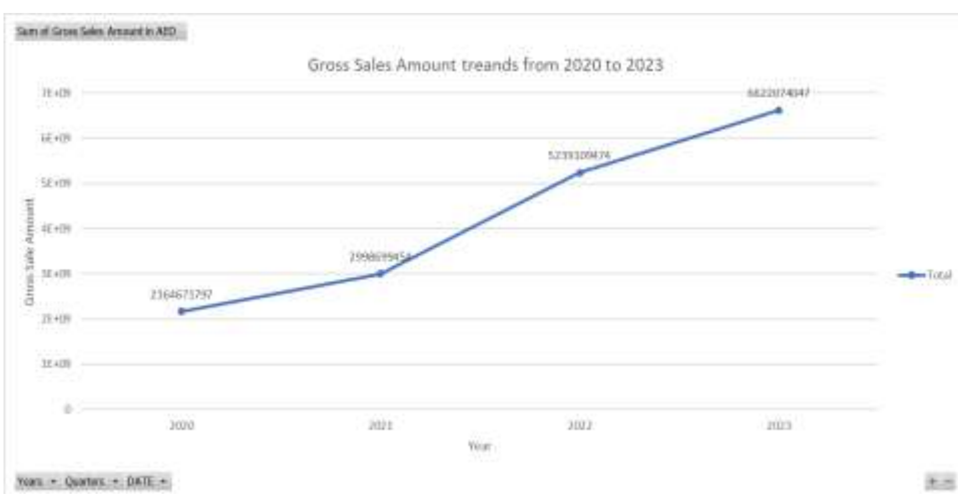

Figure 3. Time-series Analysis

The time series analysis of DL S Mall's gross sales from 2020 to 2023 shows a steady increase in customer spending. Sales began at 2.16 billion AED in 2020, likely due to the pandemic, and grew consistently to 6.62 billion AED by 2023. The largest growth occurred between 2021 and 2022, reflecting a strong post-pandemic recovery. This continued growth in 2023 indicates sustained customer engagement and positive market conditions.

Time-series Analysis showing the COVID-19 impact

The figure 4 shows the quarterly gross sales trends from 2020 to 2023, highlighting the impact of COVID-19 and the subsequent recovery.

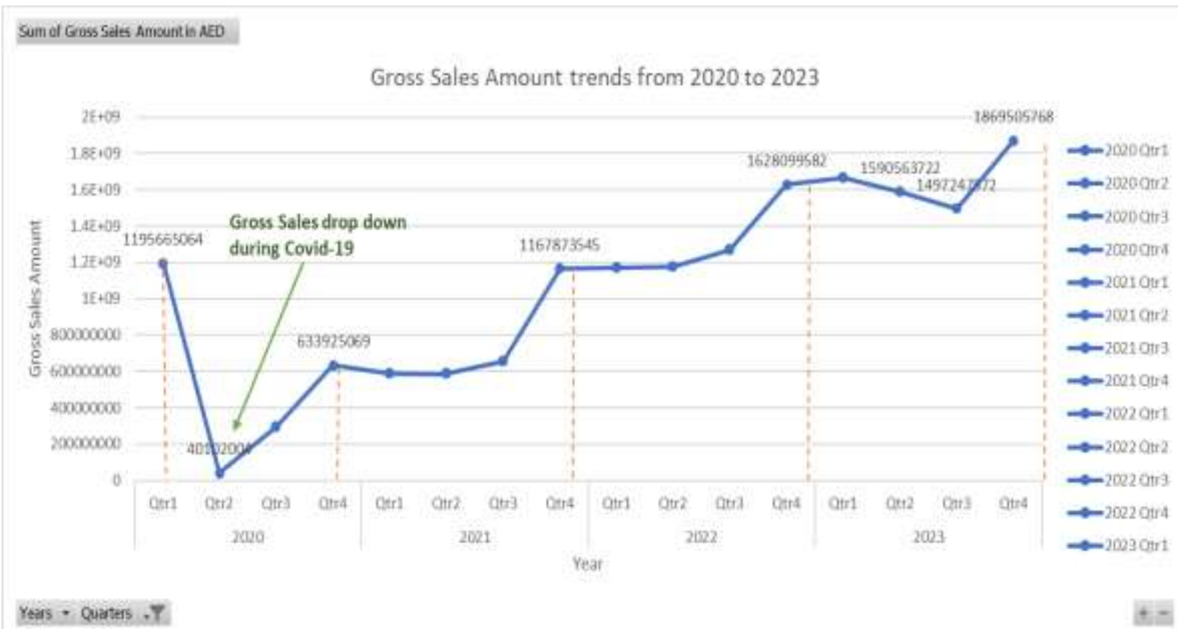

Figure 4. Time-series analysis showing the COVID-19 impact

Sales dropped sharply in Q2 2020 due to the pandemic, falling to 401 million AED, but gradually recovered, peaking in Q4 2021 at 1.17 billion AED. The growth continued into 2022, with sales reaching 1.63 billion AED by the year's end. By Q4 2023, sales hit a new high of 1.87 billion AED, indicating sustained recovery and growth. The strong sales in Q4 each year suggest this period is crucial for DLS Mall, driven by factors like holiday shopping and promotions.

Heat Map

The Figure 5 shows the quarterly gross sales trends from 2020 to 2023, highlighting the impact of COVID-19 and the subsequent recovery

Floor	Total Gross Sale Amount
Cinema Floor CF	**604101311**
CF Shops	604101311
First Floor	**4341360683**
East Wing	292566
North Wing	63731173
South Wing	1169237333
West Wing	3108099611
Ground Floor	**9235256916**
East Wing	48779281
North Wing	138862867
South Wing	1431088770
West Wing	7616525998
Second Floor	**2843835862**
East Wing	33000634
North Wing	92558204
South Wing	224996658
West Wing	2493280366

Figure 5. Heat map showing the Gross Sales Amount across different locations

The heat map highlights key sales performance areas at DL S Mall. The Ground Floor, particularly the West Wing, leads with 761.6 million AED in sales. The First Floor also performs well, driven by the West Wing, while the Cinema Floor CF and Second Floor show moderate sales, primarily in the CF Shops and West Wing. However, the East Wings and North Wings across the First and Second Floors underperform, indicating areas that need improvement to boost overall sales.

Impact of COVID-19 ON SALES and the recovery patterns

This graph in figure 6 shows the quarterly gross sales trends from 2020 to 2023, highlighting the impact of COVID-19 and the subsequent recovery.

Departments	2020	2021	2022	2023
Athletic Apparel and Gear	8602477	7659702	12082797	10959250
Bags and Leather Goods	27039955	26257228	47817876	48822601
Beauty Products	205829249	172810439	258727998	394857790
Books	1985260	2066590	4328500	4893890
Confectionary	175709954	197987834	400198139	517238899
Dubai Souvenirs	26489194	38755866	84801647	115883359
Electronic Devices	122186	49997	58632	2919
Electronic Devices and Tech	212168598	280590869	502082001	632287092
Eyewear	15145868	23960855	39526286	49846999
Fashion Boutiques	146889096	264575075	781577384	1049873056
Fashion Jewellery	8107527	8280764	20777910	20105381
Fine Jewellery	18872960	33731670	54328655	69206604
Fragrances	399863687	704059065	1130216060	1359791058
Gift Cards	23038507	49359057	15090852	7037206
Gift Items	986174	1794457	2788042	3459644
Gold	113806095	203585740	418563896	477215217
Gourmet Foods	82569099	109730910	208857884	272566985
Home Decor	32195	43929	58067	80868
Home Essentials and Toiletries	28843026	42700685	68961434	78418838
Kids Fashion and Accessories	859758	772061	1113122	1673075
Luxury Surprises	21720030	24100214	29371706	34393667
Men's Fashion and Accessories	19979248	30347299	53007972	57080657
Music Instruments	134608255	156792098	161092072	177859334
Periodicals and Office Supplies	948254	368940	1409491	1831243
Phone Cards	13094870	7514117	14201968	13858228
Photography Equipment	0	0	0	75
Smoking Products	297539863	351508867	563221229	751974383
Timepieces	144940686	210957985	268965395	357356482
Toys and Games	8260512	10055545	21065969	26596494
Travel Equipment	14432551	23529963	46617899	60876140
Women's Fashion and Accessories	9081464	10511364	19244914	15188014
Writing Instruments and Lighters	3105199	4240269	8953677	10838599

Figure 6. Heat map showing the impact of COVID-19 in departments

The heat map illustrates the impact of COVID-19 on sales from 2020 to 2023, showing a sharp decline in 2020, particularly in Home Decor, Books, and Office Supplies, as well as reductions in Fragrances, Fashion Boutiques, and Electronics. Recovery began in 2021, with strong growth in Fragrances, Confectionary, and Gourmet Foods, continuing through 2022 and 2023. Many departments, including Fragrances and Smoking Products, exceeded pre-pandemic sales levels. The appearance of Photography Equipment sales in 2023 suggests it was newly introduced or reintroduced that year.

Inventory Management

Table 1 and Figure 7 show the ABC analysis inventory analysis of DLS Mall shows that 7 departments (22%) fall under the **A category**, contributing 76% of total sales and should be prioritised for inventory management, marketing, and customer focus. The **B category** includes 9 departments (28%) contributing 20% of sales, requiring moderate stock levels to support overall revenue. Lastly, the **C category** with 16 departments (50%) contributes only 4% of sales and should be managed with minimal stock to avoid excess costs. Focusing on **A** and **B** categories will enhance inventory efficiency and profitability for DLS mall.

Table 1. ABC Analysis Results

	A	**B**	**C**	**Total**
No. of Departments	7	9	16	32
% Departments	22%	28%	50%	100%
Gross Sales Amount	12964748633	3403100512	656705627	17024554772
% Gross Sales Amount	76%	20%	4%	100%

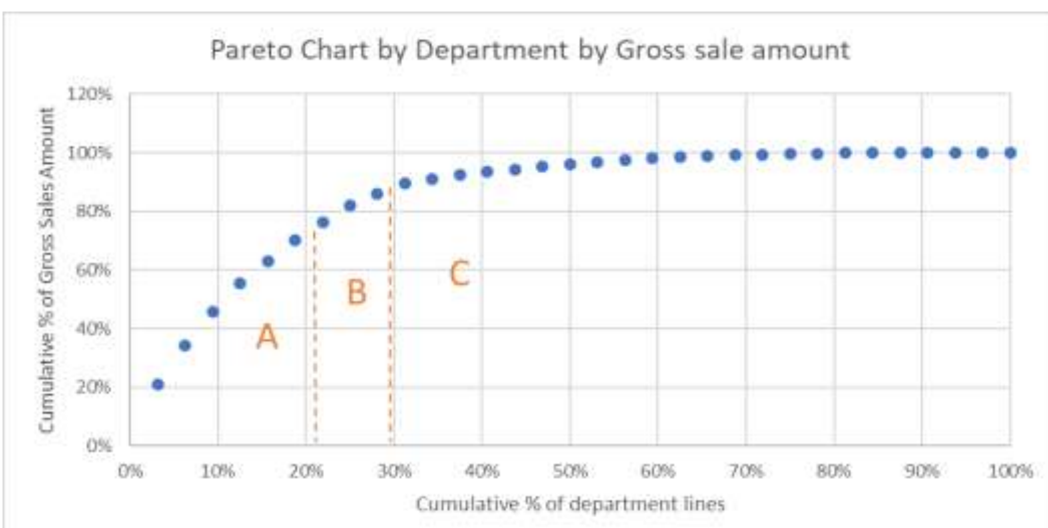

Figure 7. ABC Analysis

Forecasting future sales

The chart shown in Table 2 and Figure 8 presents the Gross Sales Amount from 2020 to 2026, showing actual data for 2020-2023 and projected sales for 2024-2026, with confidence intervals indicating the range within which sales could fall.

Table 2. Forecasting the Future Sales

Row Labels	Sum of Gross Sales Amount in AED	Forecast(Sum of Gross Sales Amount in AED)	Confidence Interval(Sum of Gross Sales Amount in AE
2020	2164671797		
2021	2998699454		
2022	5239109474		
2023	6622074047		
2024		8148517929	720939029.6
2025		9645332047	735075388.2
2026		11142146165	765827643.3

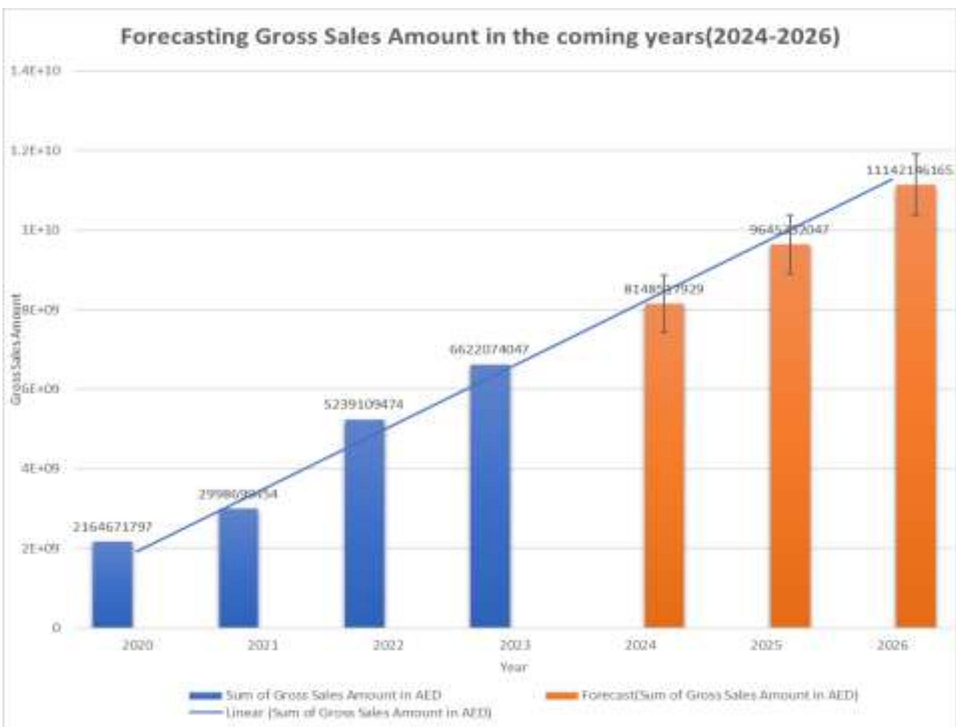

Figure 8. Forecasting Future Sales using Trendlines

Sales grew steadily from 2.16 billion AED in 2020 to 6.62 billion AED in 2023, indicating a positive trend. Forecasts predict further growth, reaching 8.15 billion AED by 2024 and 11.14 billion AED by 2026. However, potential variability requires careful risk management and flexibility in operations. To maximize this growth, the business should focus on resource allocation, expansion planning, and investing in technology and customer engagement, while remaining vigilant about external factors and market fluctuations to ensure long-term success.

Conclusion

In conclusion, the analysis of DL S Mall's sales, customer spending patterns, and inventory management revealed key insights for growth. The Ground Floor is the main revenue driver, especially in Electronic Devices and Tech and Fragrances, while the Second Floor leads in sales quantity for Confectionary and Dubai Souvenirs. The Cinema Floor CF showed lower engagement. Pareto analysis highlighted that a few departments generate most of the revenue, and time-series analysis showed steady post-COVID-19 recovery, with strong growth expected through 2026. Prioritizing inventory for high-revenue departments and optimizing resources will help ensure future profitability and competitiveness.

Future Scope

As the next step, DL S Mall should analyze customer traffic patterns to understand how footfall impacts sales, especially on the lower-performing Cinema Floor and Second Floor. Investigating the effects of marketing campaigns and promotions could also boost engagement in these areas. Further analysis of customer demographics will help identify profitable groups for targeted marketing. Additionally, collecting data on inventory turnover, customer satisfaction, and operating costs by department will improve inventory management and operational efficiency, leading to better strategic decisions for future growth.

References

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